

Ghazi Muhammad Abdullah

Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

REPORT OF TASK 2

Introduction

This report summarizes the work completed during the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The focus was on developing a strong foundation in Python programming by studying its fundamental concepts. The learning material covered Python syntax, variables, data types, strings, user input, conditional statements, and loops. These topics are essential for building programming skills and preparing for future Artificial Intelligence and Machine Learning applications.

Objectives

- Understand the fundamentals of Python programming.
- Learn how to use variables, data types, and user input.
- Explore string operations and built-in string methods.
- Understand conditional statements and decision-making.
- Learn how to use loops for repetitive tasks.

Task Description

The assigned work focused on learning the core concepts of Python programming through two modules: Python Foundations and Control Flow. The objective was to understand Python syntax, learn how programs receive and process user input, and explore decision-making and looping structures that form the basis of programming logic.

Work Done

During this work, I learned the fundamentals of Python programming, including variables, user input, data types, and basic input/output operations. I also explored string manipulation techniques such as concatenation, indexing, slicing, and commonly used string methods. In the second module, I studied control flow concepts, including **if**, **elif**, and **else** statements for decision-making, along with **for** and **while** loops for performing repetitive tasks. I also learned about the **break** and **continue** statements, which control loop execution. Overall, these modules strengthened my understanding of Python fundamentals and prepared me for future Artificial Intelligence and Machine Learning programming.

```
M-TEC Code Practice Tasks.py x
1 #Task 1.2: Take a string input. Print its first character, last character, length, and the string in uppercase.
2
3 # Take string input
4 text = input("Enter a string: ")
5 # Print first character
6 print("First character:", text[0])
7 # Print last character
8 print("Last character:", text[-1])
9 # Print length of the string
10 print("Length:", len(text))
11 # Print the string in uppercase
12 print("Uppercase:", text.upper())
13
14 #Task 2.3: Keep asking the user for numbers (while loop). Stop when they enter 0. Print the sum of all numbers entered.
15 # Initialize the sum
16 total = 0
17 # Keep asking the user for numbers
18 while True:
19     number = int(input("Enter a number (0 to stop): "))
20     if number == 0:
21         break
22     total += number
23 # Print the sum
24 print("The sum of all numbers is:", total)
25
26 #Task 3.1: Create a list of 5 student names. Print them one by one using a loop.
27 # Create a list of 5 student names
28 students = ["Ali", "Ahmed", "Ayesha", "Fatima", "Sara"]
29
30 # Print each student name using a loop
31 for student in students:
32     print(student)
```

Challenges Faced

The concepts introduced were beginner-friendly, and no major technical challenges were encountered. However, understanding the practical differences between **for** and **while** loops, along with the use of loop control statements, required careful practice. Similarly, exploring various string operations and type conversion techniques helped deepen my understanding of Python programming.

Key Learnings

This work provided a solid foundation in Python programming. I gained a clear understanding of variables, data types, user input, string manipulation, conditional statements, and loops, all of which are essential for developing efficient programs. I also learned why Python's simple and readable syntax makes it a preferred language for Artificial Intelligence and Machine Learning development. These concepts will support more advanced programming topics throughout the internship.

Conclusion

This learning experience strengthened my understanding of Python programming fundamentals and introduced the essential concepts required for Artificial Intelligence and Machine Learning development. It improved my knowledge of Python syntax, data handling, decision-making, and iterative programming. The skills gained will support future practical exercises and help me write structured, efficient, and well-organized Python programs during the remainder of the internship.

Intern Name: Ghazi Muhammad Abdullah

Supervisor Name: Mr. Mohazin Zahid

Date: June 30th, 2026

Domain: AI/ML Intern

Intern Signature:

A handwritten signature in black ink, appearing to read 'Ghazi', with a stylized flourish at the end.